

Medical Insurance Charges Analysis

Data Science Capstone Project

Exploratory Analysis • SQL • Linear Regression

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Business Problem & Objective

Business problem

- Understand how patient characteristics relate to medical insurance charges.
- Support more consistent, data-informed cost estimation.

Objective

- Explore age, sex, BMI, children, smoking status and region.
- Build a Linear Regression model to predict charges.
- Translate findings into business insights and recommendations.

Understand Data



Explore Patterns



Build Model



Business Insights

Dataset & Data Preparation

Clean data, but with one important lesson about outliers

1,338

Original observations

7

Columns

0

Missing values

1

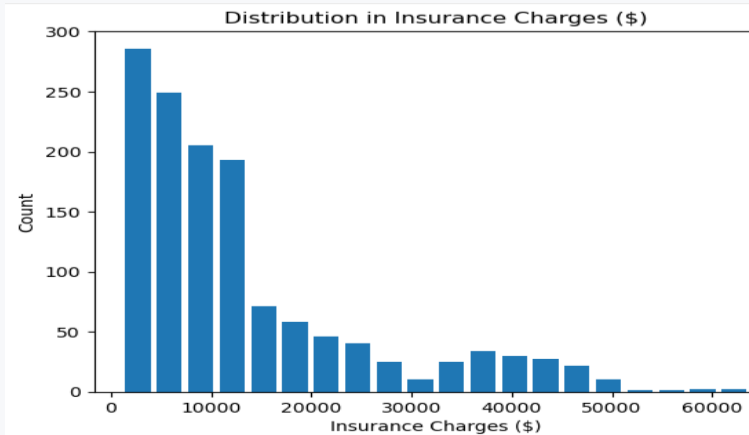
Exact duplicate removed

Outlier investigation preserved valid information

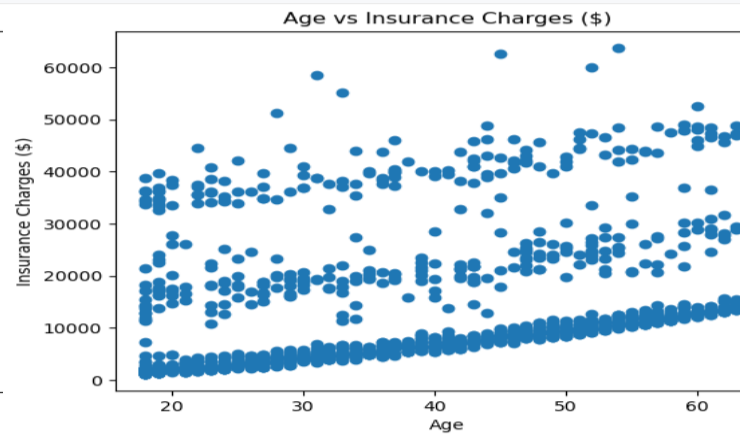
- Potential outliers were investigated rather than automatically deleted.
- Some high charges that looked unusual overall were reasonable within the smoker category.
- Context mattered: group-level investigation helped protect data integrity.

What Did EDA Reveal?

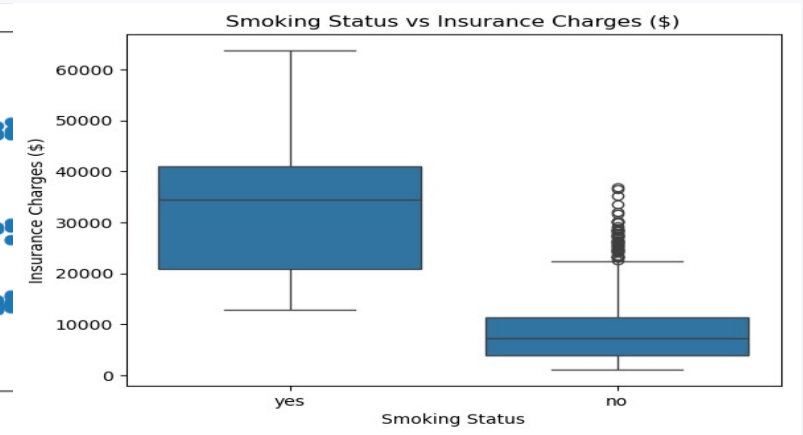
Three visuals capture the main story



Charges are right-skewed



Charges generally rise with age

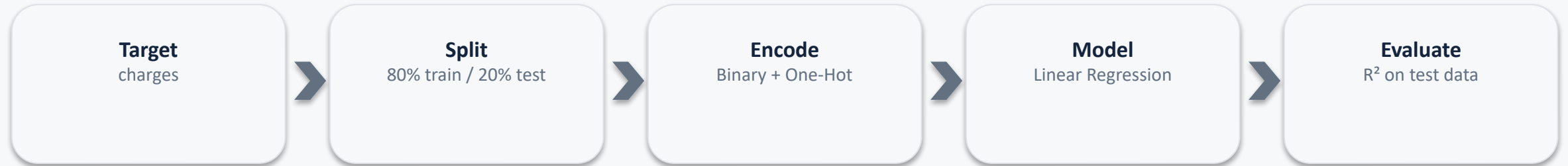


Smoking shows the clearest separation

Takeaway: Smoking status showed the strongest visual difference; age had a positive relationship with charges, while BMI was weaker when examined independently.

From Data to Prediction

How the Linear Regression model was built and evaluated



$$R^2 = 0.807$$

≈ 80.7% of the variation in test-set insurance charges was explained by the model

What Did the Model Learn?

Interpreting the model coefficients while holding other variables constant

+\$23,077.76

Smoker vs non-smoker

Largest estimated difference

+\$533.01

Each additional child

+\$318.70

Each BMI unit

+\$248.21

Each additional year

Sex

Male vs female: -\$101.54

Region vs Northeast (reference)

Northwest -\$391.76 | Southeast -\$838.92 | Southwest -\$659.14

Interpretation note: coefficients describe associations in this fitted model, not causal effects.

SQL Insights: Another View of the Data

SQL summaries provided another way to compare insurance charges

\$32,050

Average charge — smokers

\$8,434

Average charge — non-smokers

≈ 3.8×

Smoker/non-smoker average

\$21,248

Highest age-group average

Age 60+

\$15,552

Highest BMI-category average

Obesity

\$14,735

Highest regional average

Southeast

Business Recommendations & Conclusion

1

Use multiple characteristics together

Avoid relying on a single patient attribute when estimating charges.

2

Give smoking status appropriate consideration

It showed a strong, consistent association with higher charges across EDA, SQL and the model.

3

Combine modelling with exploratory monitoring

EDA reveals distributions, group differences and unusual observations that one score cannot fully communicate.

Conclusion: The model explained $\approx 80.7\%$ of test-set charge variation, while smoking status emerged as one of the clearest characteristics associated with higher charges.